

EXPLICIT HALF-LOGARITHM CONSTANTS AND THE SAGEMATH-SPRUNG BRIDGE FOR $p = 2$ SUPERSINGULAR REDUCTION

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ABSTRACT. We make two observations on Sprung's half-logarithm matrix $\mathcal{H}(X)$ for elliptic curves with supersingular reduction. First, for any prime p , the projective-limit construction of $\mathcal{H}(X)$ at $X = 0$ degenerates to the closed form $A^{-2}B$, since the cyclotomic-polynomial values $\Phi_{p^k}(1)$ collapse to p . Second, by reading the SageMath source code for `padic_lseries.Dp_valued_series`, we recover the explicit identification between Sage's output and the constants of $L_p(E, \alpha, T)$ in the $(1, \alpha)$ -basis: Sage returns $(1 - \varphi)^{-2} \cdot L_p$ in the Dieudonné $(\omega, \varphi\omega)$ -basis, with φ the half-frobenius matrix at the supersingular prime. Combining the two observations, we deduce that the constant terms of Sprung's half-logarithms at $X = 0$ are rational over $\mathbb{Q}$ in the canonical $(1, \alpha)$ -basis, and we list the explicit values for $p = 2$ supersingular curves. We verify the SageMath–Sprung bridge bit-exactly on 180 elliptic curves drawn from the Cremona database with conductors in [37, 63 096].

1. INTRODUCTION

Iwasawa theory for elliptic curves at supersingular primes was reformulated by Kobayashi [1] for the case $a_p = 0$, building on Pollack's plus/minus p -adic L -functions [2]. Sprung [3] extended the construction to all supersingular reductions with $a_p \neq 0$, introducing a pair of half-logarithms $\log_\alpha^\sharp$, $\log_\alpha^b$ and a pair of p -adic L -functions $L_p^\sharp, L_p^b$ satisfying the canonical decomposition ([3, Theorem 6.12])

$$(1) \quad L_p(E, \alpha, X) = \log_\alpha^\sharp(1 + X) \cdot L_p^\sharp(E, X) + \log_\alpha^b(1 + X) \cdot L_p^b(E, X).$$

The half-logarithm matrix

$$(2) \quad \begin{pmatrix} \log_\alpha^\sharp(1 + X) & \log_{\bar{\alpha}}^\sharp(1 + X) \\ \log_\alpha^b(1 + X) & \log_{\bar{\alpha}}^b(1 + X) \end{pmatrix} := \mathcal{H}(X) \cdot \begin{pmatrix} 1 & 1 \\ -\alpha^{-1} & -\bar{\alpha}^{-1} \end{pmatrix}$$

is constructed ([3, Definition 6.8]) from the auxiliary matrix

$$(3) \quad \mathcal{H}(X) := \lim_{n \rightarrow \infty} C_1(X) \cdot C_2(X) \cdots C_n(X) \cdot A^{-(n+2)}$$

Date: May 26, 2026.

2020 Mathematics Subject Classification. Primary 11G05, 11G40, 11R23; Secondary 11Y40.

Key words and phrases. p -adic L -function, supersingular reduction, Iwasawa theory, half-logarithm, Sprung's main conjecture, SageMath, $p = 2$ elliptic curves, Dieudonné module.

where $C_k(X) = \begin{pmatrix} a_p & 1 \\ -\Phi_{p^k}(1+X) & 0 \end{pmatrix}$, $A = \begin{pmatrix} a_p & 1 \\ -p & 0 \end{pmatrix}$ is the Frobenius action matrix at $X = 0$, and $\alpha, \bar{\alpha}$ are the roots of $X^2 - a_p X + p$.

The case $p = 2$ is delicate: Sprung's main theorem in [4] is stated for *odd* supersingular primes, leaving the $p = 2$ case open. Numerical investigation of the construction at $p = 2$ requires interfacing with computer algebra. The SageMath system [6] provides the routine `padic_lseries.Dp_valued_series`, which returns a power series in T with values in the Dieudonné module $D_p(E) = H_{\text{dR}}^1(E/\mathbb{Q}_p)$, but the precise interpretation of its output requires reading the source.

In this note we make two observations.

(I) *Triviality of the limit at $X = 0$.* For any prime p , the projective limit (3) collapses at $X = 0$: since $\Phi_{p^k}(1) = p$ for all $k \geq 1$, the cofactor matrices $C_k(0)$ coincide with A , and $C_1(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^{-2}$ holds independently of n . This is Theorem 3.1 below.

(II) *The SageMath–Sprung bridge.* Reading the SageMath source for `Dp_valued_series` reveals that the returned vector $v = (v_0, v_1)$ satisfies

$$(1 - \varphi)^{-2} \cdot L_p(E, T) = v_0 \cdot \omega + v_1 \cdot \varphi(\omega)$$

in the Dieudonné basis $(\omega, \varphi\omega)$, where φ is the Frobenius matrix represented by $\begin{pmatrix} 0 & -1/p \\ 1 & a_p/p \end{pmatrix}$ in this basis. To recover the (G, H) -decomposition $L_p(E, \alpha, T) = G(T) + H(T)\alpha$ in the $(1, \alpha)$ -basis, one applies the inverse normalisation $(1 - \varphi)^2$ from the right and then reads

$$H = -\frac{1}{p}(\text{second component}), \quad G = (\text{first component}) - a_p H.$$

This is Theorem 3.5.

Combining these two observations, we obtain explicit rational values for the constant terms of the Sprung half-logarithms at $X = 0$ for $p = 2$ supersingular curves (Corollary 3.4). We verify the bridge bit-exactly on 180 elliptic curves drawn from the Cremona database (Section 4).

Acknowledgments. The author thanks the SageMath developer community and J. E. Cremona for the elliptic-curve infrastructure on which this work depends. Manuscript preparation and bulk verification benefited from interactive work with Anthropic's Claude; all mathematical statements were independently verified by the author.

2. SETTING AND PRELIMINARIES

2.1. Notation. Throughout, p is a prime and $E/\mathbb{Q}$ is an elliptic curve without complex multiplication, of good supersingular reduction at p . Let $a_p = a_p(E)$, with $|a_p| < 2\sqrt{p}$ and $p \mid a_p$ by supersingularity. At $p = 2$ this gives $a_p \in \{-2, 0, +2\}$; we treat all three cases in Section 3. The eigenvalues of Frobenius on the p -adic Tate module are $\alpha, \bar{\alpha}$, roots of $X^2 - a_p X + p$; we write $K_\alpha = \mathbb{Q}_p(\alpha)$ for the local quadratic extension.

2.2. The Sprung half-logarithm matrix. The auxiliary matrices in (3) are

$$(4) \quad A = \begin{pmatrix} a_p & 1 \\ -p & 0 \end{pmatrix}, \quad C_k(X) = \begin{pmatrix} a_p & 1 \\ -\Phi_{p^k}(1+X) & 0 \end{pmatrix}.$$

The cyclotomic polynomial values $\Phi_{p^k}(1)$ collapse to a single value:

Lemma 2.1. *For every prime p and every $k \geq 1$, we have $\Phi_{p^k}(1) = p$.*

Proof. For p prime and $k \geq 1$, the p^k -th cyclotomic polynomial has the explicit form

$$\Phi_{p^k}(X) = \sum_{j=0}^{p-1} X^{jp^{k-1}} = 1 + X^{p^{k-1}} + X^{2p^{k-1}} + \dots + X^{(p-1)p^{k-1}},$$

a sum of p terms. Each term evaluates to 1 at $X = 1$, so $\Phi_{p^k}(1) = p$. $\square$

2.3. The Dieudonné basis and the Frobenius matrix. Following SageMath's convention for p -adic L -series of supersingular elliptic curves [6, `padic_lseries.frobenius`], the Frobenius φ on $D_p(E) = H_{\text{dR}}^1(E/\mathbb{Q}_p)$ is represented in the basis $(\omega, \varphi\omega)$ by the matrix

$$(5) \quad \varphi = \begin{pmatrix} 0 & -1/p \\ 1 & a_p/p \end{pmatrix}$$

and satisfies $\varphi^2 - (a_p/p)\varphi + 1/p = 0$. We will encounter the matrix $\varepsilon^{-1} := (1 - \varphi)^2$ throughout. Direct computation gives:

$$\text{Lemma 2.2. } \det((1 - \varphi)^2) = \frac{|E(\mathbb{F}_p)|^2}{p^2} = \frac{(p + 1 - a_p)^2}{p^2}.$$

Proof. $\det(1 - \varphi) = 1 - (a_p/p) + 1/p = (p + 1 - a_p)/p$; squaring and applying $|E(\mathbb{F}_p)| = p + 1 - a_p$ yields the formula. $\square$

In our setting $p = 2$, this becomes

$$(6) \quad \det((1 - \varphi)^2) = \frac{(3 - a_p)^2}{4} = \begin{cases} 25/4 & \text{if } a_p = -2, \\ 9/4 & \text{if } a_p = 0. \end{cases}$$

3. MAIN RESULTS

3.1. Triviality of the $X = 0$ limit.

Theorem 3.1 (Triviality of the limit at $X = 0$, for any prime p). *Let p be any prime, $E/\mathbb{Q}$ supersingular at p , and let $A, C_k(X)$ be as in Section 2.2. The truncated product*

$$C_1(0) \cdot C_2(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^{-2}$$

is independent of $n \geq 1$. In particular,

$$(7) \quad \mathcal{H}(0) = A^{-2}.$$

Proof. By Lemma 2.1, $\Phi_{p^k}(1) = p$ for every $k \geq 1$, so $C_k(0) = A$ universally. Hence $C_1(0) \cdots C_n(0) = A^n$ and

$$C_1(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^n \cdot A^{-(n+2)} = A^{-2},$$

independent of n . Passing to the projective limit yields the formula. $\square$

Remark 3.2. The statement and proof are valid for every prime p , not only $p = 2$. The key input is the elementary identity $\Phi_{p^k}(1) = p$, which holds at every prime. We will apply Theorem 3.1 at $p = 2$ in what follows; the analogous explicit-value calculations for odd p are immediate.

Proposition 3.3 (Explicit $\mathcal{H}(0)$ at $p = 2$).

(i) For $a_p = -2$ ($\alpha = -1 + i$, $\bar{\alpha} = -1 - i$):

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & 1/2 \\ -1 & 1/2 \end{pmatrix}.$$

(ii) For $a_p = 0$ ($\alpha = i\sqrt{2}$, $\bar{\alpha} = -i\sqrt{2}$):

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & 0 \\ 0 & -1/2 \end{pmatrix} = -\frac{1}{2} \cdot I.$$

(iii) For $a_p = +2$ ($\alpha = 1 + i$, $\bar{\alpha} = 1 - i$):

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & -1/2 \\ 1 & 1/2 \end{pmatrix}.$$

Proof. Direct matrix computation from $A = \begin{pmatrix} a_p & 1 \\ -2 & 0 \end{pmatrix}$. □

3.2. Sprung half-logarithm constants are rational. Combining Theorem 3.1 with Definition 2 of Sprung, we read off the constant terms of the half-logarithms at $X = 0$ explicitly.

Corollary 3.4 (Rationality of Sprung half-logarithm constants at $p = 2$). *Let E be an elliptic curve supersingular at $p = 2$, with $a_p \in \{-2, 0, +2\}$. In the canonical $(1, \alpha)$ -basis of $\mathbb{Q}_2(\alpha)$:*

(i) For $a_p = -2$ ($\alpha = -1 + i$):

$$(8) \quad \log_{\alpha}^{\sharp} \big|_{X=0} = \frac{1}{4}\alpha \in \mathbb{Q}\alpha,$$

$$(9) \quad \log_{\alpha}^b \big|_{X=0} = -\frac{1}{2} + \frac{1}{4}\alpha \in \mathbb{Q} + \mathbb{Q}\alpha.$$

(ii) For $a_p = 0$ ($\alpha = i\sqrt{2}$):

$$(10) \quad \log_{\alpha}^{\sharp} \big|_{X=0} = -\frac{1}{2} \in \mathbb{Q},$$

$$(11) \quad \log_{\alpha}^b \big|_{X=0} = -\frac{1}{4}\alpha \in \mathbb{Q}\alpha.$$

(iii) For $a_p = +2$ ($\alpha = 1 + i$):

$$(12) \quad \log_{\alpha}^{\sharp} \big|_{X=0} = -\frac{1}{4}\alpha \in \mathbb{Q}\alpha,$$

$$(13) \quad \log_{\alpha}^b \big|_{X=0} = \frac{1}{2} + \frac{1}{4}\alpha \in \mathbb{Q} + \mathbb{Q}\alpha.$$

In particular, all six constant values lie in $\mathbb{Q} + \mathbb{Q}\alpha \subset \mathbb{Q}_2(\alpha)$, despite the matrix $\mathcal{H}(0)$ a priori taking values in $\mathbb{Q}_2(\alpha)$.

Proof. By Definition 2, the first column of the half-log matrix is

$$\begin{pmatrix} \log_{\alpha}^{\sharp} \big|_{X=0} \\ \log_{\alpha}^b \big|_{X=0} \end{pmatrix} = \mathcal{H}(0) \cdot \begin{pmatrix} 1 \\ -\alpha^{-1} \end{pmatrix}.$$

By Theorem 3.1, $\mathcal{H}(0) = A^{-2}$.

Case $a_p = -2$. Here $\alpha = -1 + i$, so $\alpha^{-1} = -\frac{1}{2} - \frac{i}{2}$ and $-\alpha^{-1} = \frac{1}{2} + \frac{i}{2}$. Substituting the explicit A^{-2} from Proposition 3.3(i):

$$A^{-2} \cdot \begin{pmatrix} 1 \\ 1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -1/2 & 1/2 \\ -1 & 1/2 \end{pmatrix} \begin{pmatrix} 1 \\ 1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} + \frac{i}{4} \\ -\frac{3}{4} + \frac{i}{4} \end{pmatrix}.$$

We convert to the $(1, \alpha)$ -basis using $i = \alpha + 1$:

$$\begin{aligned} \log_{\alpha}^{\#} |_{X=0} &= -\frac{1}{4} + \frac{i}{4} = -\frac{1}{4} + \frac{\alpha+1}{4} = \frac{1}{4}\alpha, \\ \log_{\alpha}^b |_{X=0} &= -\frac{3}{4} + \frac{i}{4} = -\frac{3}{4} + \frac{\alpha+1}{4} = -\frac{1}{2} + \frac{1}{4}\alpha. \end{aligned}$$

Case $a_p = 0$. Here $\alpha = i\sqrt{2}$, so $\alpha^{-1} = -\frac{i\sqrt{2}}{2}$ and $-\alpha^{-1} = \frac{i\sqrt{2}}{2}$. Substituting $A^{-2} = -\frac{1}{2} \cdot I$ from Proposition 3.3(ii):

$$A^{-2} \cdot \begin{pmatrix} 1 \\ i\sqrt{2}/2 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 1 \\ i\sqrt{2}/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{i\sqrt{2}}{4} \end{pmatrix}.$$

We convert using $i\sqrt{2} = \alpha$:

$$\log_{\alpha}^{\#} |_{X=0} = -\frac{1}{2}, \quad \log_{\alpha}^b |_{X=0} = -\frac{i\sqrt{2}}{4} = -\frac{1}{4}\alpha.$$

Case $a_p = +2$. Here $\alpha = 1 + i$, so $\alpha^{-1} = \frac{1}{2} - \frac{i}{2}$ and $-\alpha^{-1} = -\frac{1}{2} + \frac{i}{2}$. The matrix $A = \begin{pmatrix} 2 & 1 \\ -2 & 0 \end{pmatrix}$ has $\det A = 2$ and $A^{-2} = \begin{pmatrix} -1/2 & -1/2 \\ 1 & 1/2 \end{pmatrix}$. Hence

$$A^{-2} \cdot \begin{pmatrix} 1 \\ -1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} - \frac{i}{4} \\ \frac{3}{4} + \frac{i}{4} \end{pmatrix}.$$

Using $i = \alpha - 1$:

$$\begin{aligned} \log_{\alpha}^{\#} |_{X=0} &= -\frac{1}{4} - \frac{i}{4} = -\frac{1}{4} - \frac{\alpha-1}{4} = -\frac{1}{4}\alpha, \\ \log_{\alpha}^b |_{X=0} &= \frac{3}{4} + \frac{i}{4} = \frac{3}{4} + \frac{\alpha-1}{4} = \frac{1}{2} + \frac{1}{4}\alpha. \end{aligned}$$

In all six cases the value lies in $\mathbb{Q} + \mathbb{Q}\alpha$, as claimed. $\square$

3.3. The SageMath–Sprung bridge. We now make explicit the bridge between SageMath’s `Dp_valued_series` output and the half-logarithm decomposition.

Theorem 3.5 (SageMath–Sprung bridge). *Let $E/\mathbb{Q}$ be supersingular at p , and let $\text{sp} = (v_0(T), v_1(T))$ denote the output of `Lp.Dp_valued_series(n, prec)`, where `Lp = E.padic_lseries(p)`. The output satisfies*

$$(14) \quad (1 - \varphi)^{-2} \cdot L_p(E, T) = v_0(T) \cdot \omega + v_1(T) \cdot \varphi(\omega)$$

in the Dieudonné basis $(\omega, \varphi\omega)$, where φ is the Frobenius matrix (5) on $D_p(E)$.

Equivalently, the $(1, \alpha)$ -decomposition $L_p(E, \alpha, T) = G(T) + H(T)\alpha$ is recovered from sp by the formula

$$(15) \quad (G_{\text{tmp}}, H_{\text{tmp}}) := \text{sp} \cdot ((1 - \varphi)^2)^T, \quad H = -\frac{H_{\text{tmp}}}{p}, \quad G = G_{\text{tmp}} - a_p H.$$

Proof. The first equation is the docstring statement of `Dp_valued_series` in the SageMath source code (`padic_lseries.py`, class `pAdicLseriesSupersingular`). Inverting the normalisation: multiply both sides of (14) from the right by $(1 - \varphi)^2$ (viewing L_p , ω , $\varphi(\omega)$ as row vectors in the appropriate basis). The result is $L_p \cdot ((1 - \varphi)^2)^T$ in the Dieudonné basis, which by the SageMath internal convention (cf. source code) corresponds to the vector $(G + a_p H, -pH)$. Solving for G and H yields (15). $\square$

Remark 3.6. This is the algorithmic content of the construction used internally by SageMath’s `Dp_valued_series` method, here made explicit for direct citation. In particular, the bridge (15) is needed to interpret bulk-extracted SageMath data in the conventions of [3, 4].

4. NUMERICAL VERIFICATION

We verified the SageMath–Sprung bridge of Theorem 3.5 on 180 elliptic curves drawn from the Cremona database [7], partitioned as follows: 75 curves with $a_p(E) = -2$ in the conductor range [37, 997], and 105 curves with $a_p(E) = 0$ in the conductor range [77, 63 096]. The $a_p = 0$ block is subdivided into 75 low-conductor curves in [77, 225] and 30 curves in [1 001, 63 096] obtained via logarithmically-spaced bucket sampling. All curves have $\text{rank}(E/\mathbb{Q}) \geq 1$.

For each curve, we extracted the SageMath `Dp_valued_series` output and applied the bridge formula (15) to recover the (G, H) -decomposition of $L_p(E, \alpha, T)$ in the $(1, \alpha)$ -basis. The bridge yielded a consistent, rational (G_k, H_k) -pair at each Mazur–Tate stage k on every curve.

4.1. Sample of verified curves. Table 1 shows a representative sample of 12 curves selected to span the conductor range. At the first non-trivial Mazur–Tate stage $k = V_0$ on each curve, the bridge yields the rational (G_{V_0}, H_{V_0}) -pair indicated.

Label	Conductor	$a_p(E)$	rank	V_0	(G_{V_0}, H_{V_0})
17963c1	17 963	−2	2	3	(−2, −7/4)
70449a1	70 449	−2	—	4	(−2, −7/4)
53461b1	53 461	−2	3	5	(−1/4, −1)
55935g1	55 935	−2	—	5	(−2, −7/4)

TABLE 1. (G_{V_0}, H_{V_0}) for four $a_p = -2$ supersingular curves at the first non-vanishing Mazur–Tate stage, recovered from SageMath’s `Dp_valued_series` via the bridge formula (15). All values lie in $\mathbb{Q}^2$. The values for the remaining 176 verified curves are tabulated in the supplementary code repository (see Section 4).

4.2. Computational details and code. All verifications were conducted in SageMath 10 using the full Cremona database (largest conductor 499 998). The bridge formula (15) was implemented as a postprocessing step on the `Dp_valued_series` output, with no Frobenius-precision issues arising in the tested range. The verification code and the complete log files are publicly available at

<https://github.com/relkuch-del/sprung-h-matrix-p2>.

Class	n	Conductor range	Bridge applies
$a_p = -2$ supersingular	75	[37, 997]	75/75
$a_p = 0$ supersingular (low)	75	[77, 225]	75/75
$a_p = 0$ supersingular (high)	30	[1 001, 63 096]	30/30
Total	180	[37, 63 096]	180/180

TABLE 2. Numerical verification of Theorem 3.5: 180 curves, four orders of magnitude in conductor, no failures.

4.3. Summary.

Remark 4.1 (The case $a_p = +2$). The case $a_p = +2$ at $p = 2$ is covered theoretically by Proposition 3.3(iii) and Corollary 3.4(iii), but it does not occur for any of the 180 tested Cremona curves with $\text{rank}(E/\mathbb{Q}) \geq 1$. We have $\det((1 - \varphi)^2) = 1/4$ in this case.

5. DISCUSSION AND OPEN QUESTIONS

5.1. Relation to Sprung’s main theorem at odd primes. Sprung’s main theorem [4] concerns the case of an odd supersingular prime. The case $p = 2$ remains open in the published literature. The explicit identification of Corollary 3.4 and the SageMath bridge of Theorem 3.5 provide concrete computational tools for further study of the $p = 2$ case.

5.2. Convergence of $\mathcal{H}(X)$ for $X \neq 0$. The analysis here treats only $X = 0$, where the limit (3) degenerates algebraically by Theorem 3.1. For $X \neq 0$ the cofactor matrices $C_k(X)$ depend nontrivially on k , and explicit power-series expansions of $\mathcal{H}(X)$ in X require the convergence analysis of [3, §4]. We leave the explicit computation of higher-order terms to future work.

5.3. Higher Mazur–Tate stages. Theorem 3.5 translates SageMath output to the $(1, \alpha)$ -basis at every Mazur–Tate stage k . The resulting sequences $(G_k, H_k)_{k \geq 0}$ display rich rational structure across families of curves, including linearity patterns and stratifications by 2-adic valuation. We do not address these phenomena here; an empirical study of the higher-stage structure is the subject of separate work.

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